

Dynamic Capacitance (CDYN)

Compiled By : Sujit Pant

How GT sort measures it, why it matters for high-performance silicon,
and device-Ceff correlation notes — in plain language

Contents

- **0. Why CDYN is critical for high-performance devices** — power, DVFS, and the frequency wall.
- **1–3. The CDYN method** — the big idea, the recipe, and the detailed calculation.
- **4. Why two voltages** — healthy vs. suspect dies.
- **5–6. Real results & verification** — 1,439-die population and 1439/1439 math match.
- **Appendix A. Device Ceff from e-test** — RA4/RJ4/RK4, Td→frequency, units.
- **Appendix B. CDYN vs Ceff correlation** — IDW method and the device- vs BEOL-limited finding.

0. Why CDYN is critical for high-performance devices

Dynamic capacitance (CDYN) is one of the single most important electrical health metrics for a high-performance CPU/GPU, because it sits at the center of the equation that governs how much power the chip burns while switching:

$$P_{\text{dyn}} = C_{\text{dyn}} \times V_{\text{cc}}^2 \times f \times \alpha$$

Here is why that makes CDYN mission-critical for a leading-edge part:

- **It sets dynamic power — the #1 limiter of performance.** Every extra bit of switched capacitance costs power on every clock edge. In a high-frequency device running billions of cycles per second, even a small CDYN increase translates into watts of extra heat, which directly caps the achievable frequency (thermal/power budget).
- **It drives the frequency-vs-power trade (DVFS).** Modern chips constantly move between voltage/frequency states. Because power scales with $C_{\text{dyn}} \cdot V^2 \cdot f$, an accurate CDYN per rail is what lets the power-management firmware pick the highest safe frequency at each voltage. Bad CDYN data = conservative (slow) or unsafe (hot) operating points.
- **It gates battery life and TDP compliance.** For the product to hit its thermal design power and efficiency targets, CDYN must be characterized and screened — parts with abnormally high CDYN would exceed the power envelope at target frequency.
- **It is a process-health early-warning.** CDYN reflects the real switched load (device + wire + clock + array). A drift in CDYN across a lot flags a process excursion (e.g., BEOL capacitance shift) long before it shows up as a yield or reliability problem.
- **It protects against silent bin loss.** A mis-measured or outlier CDYN can push a good die into a fail bin (bin 253). Robust CDYN extraction with outlier gating keeps healthy high-performance parts from being needlessly scrapped.

In short: for a high-performance device, **performance is power-limited, and power is CDYN-limited.** Getting CDYN right — accurately measured, correctly calculated, and screened per operating voltage — is what allows the part to run as fast as possible within its thermal and reliability budget. That is why the rest of this document focuses on exactly how GT sort computes CDYN, how we verified it, and how it relates to device-level capacitance from e-test.

1. The big idea — what is CDYN?

Every time the chip's transistors switch, they charge and discharge tiny capacitors. CDYN is a single number that captures 'how much capacitance the chip switches each clock cycle.' Water analogy:

frequency = how often you flush; **CDYN** = tank size; **current (DICC)** = the flow you measure.

Measure the current at several frequencies, see how fast it grows, and that growth rate (slope) reveals the tank size.

The one-line formula:

$$\text{CDYN (nF)} = 1000 \div \text{Voltage} \times \text{slope(current vs. frequency)}$$

2. The recipe — what the test program does

Measure DICC at three frequencies (200/300/400 MHz) at two rails (0.7 V, 0.95 V), then:

#	Step	Plain English
1	Collect	Read DICC at 200/300/400 MHz (low/mid/high).
2	Fit a line	Best straight line through the 3 points.
3	Take slope	Extra current per extra MHz = heart of CDYN.
4	Divide by V	Slope ÷ supply voltage recovers capacitance.
5	Fix units	×1000 so answer is in nanofarads (nF).
6	Round & guard	Round 4 dp; reject outside 0.01–200 nF.
7	Log	Print & pack into ^GT:31.16%0.7%19.15.

3. The detailed calculation

Physics: **Current = CDYN × Voltage × Frequency**. Hold voltage fixed, sweep frequency, and the current rises in a straight line whose slope = CDYN × Voltage. So CDYN = slope ÷ Voltage.

The constants:

- **×1000**: unit bookkeeping (MHz vs nF).
- **÷0.7 and ÷0.95**: the two test voltages (the '÷ Voltage' step).
- **+1e-8**: a microscopic never-zero nudge; too small to matter.

Worked example — first die (EFUSE_Q615S1BA_009_-10_-1)

Rail	200 MHz	300 MHz	400 MHz	Slope (A/MHz)	CDYN = 1000 × slope ÷ V
V1 = 0.7 V	5.6935 A	7.7046 A	10.0554 A	0.021810	$1000 \times 0.021810 \div 0.7 =$ 31.16 nF
V2 = 0.95 V	8.8179 A	11.6368 A	14.8594 A	0.030208	$1000 \times 0.030208 \div 0.95 =$ 31.80 nF

Tester logged **^GT:31.1564%0.7%19.15** — matches our by-hand answer exactly.

One real die (EFUSE_Q615S1BA_009_-10_-1): the actual 3-point fit

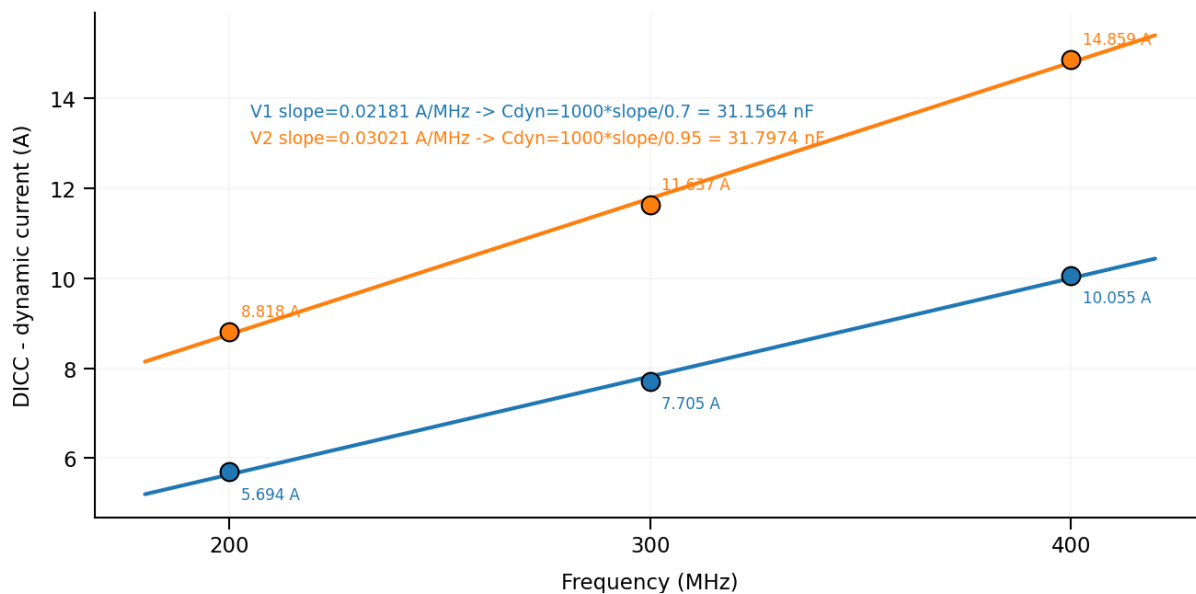


Figure 1. The actual 3-point fit for this real die. Steeper orange (0.95 V) = more current per MHz.

How CDYN (dynamic capacitance) is calculated - DICC slope / Vcc

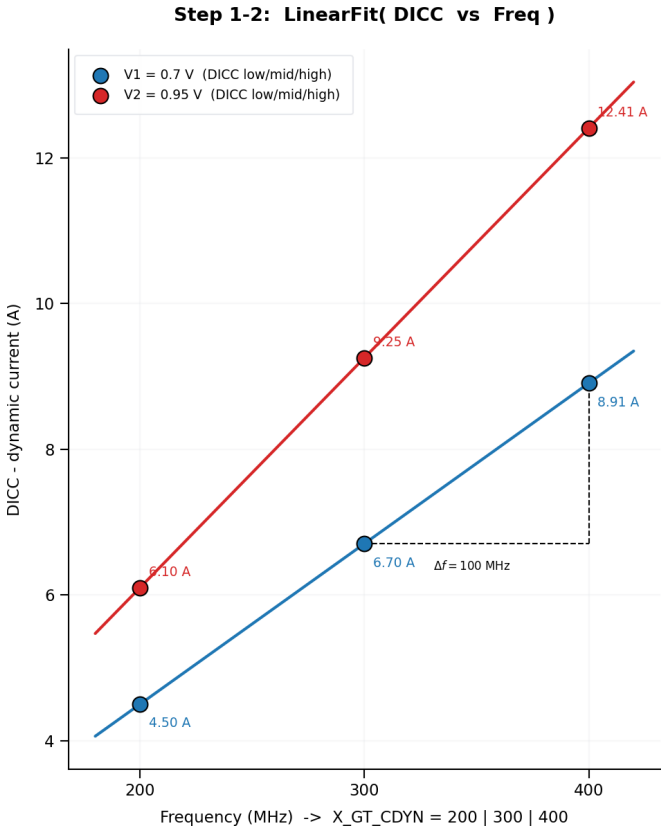
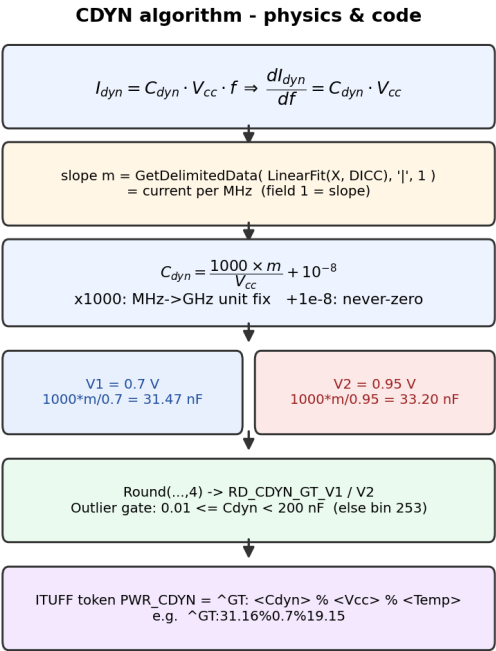


Figure 2. The whole algorithm: fit the slope, divide by voltage, fix units, gate, log.

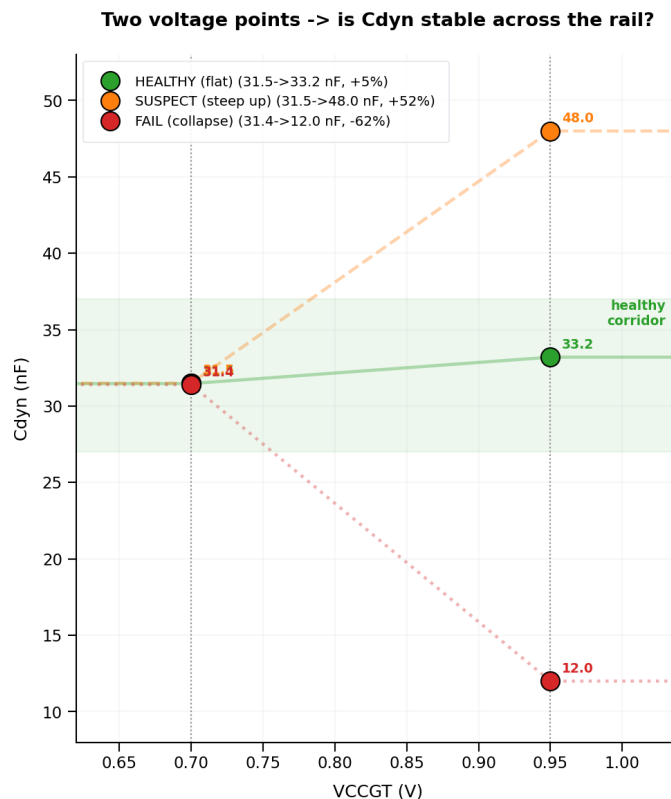


4. Why measure at two voltages?

- **Real silicon isn't a textbook capacitor** — gate and junction caps shift a little with voltage.
- **The power model needs CDYN at real operating voltages** (DVFS), so we anchor near where the chip runs.
- **Free health check** — two independent numbers flag a bad die (jump or collapse across rails).

Distinction: the 3-frequency sweep separates real switching current from leakage; the 2-voltage sweep checks CDYN stability across the operating range.

Cdyn vs Vcc - why the flow takes two voltage points



Why measure Cdyn at 2 voltages?

(1) Cdyn is mildly V-dependent (real silicon)
gate C-V + junction caps shift with bias

(2) Power model needs Cdyn at each DVFS point
 $P_{dyn} = C_{dyn} \cdot V_{cc}^2 \cdot f$

(3) Free consistency / screening check
flat=healthy big delta=marginal/bad fit

GREEN flat -> healthy
ORANGE steep up -> suspect
RED collapse -> fail

3 FREQ (200/300/400) -> Cdyn (slope) vs leakage
2 VOLTAGE (0.7/0.95) -> stability check

Figure 3. Healthy (flat), suspect (steep), and fail (collapse) signatures across voltage. Real dies are the flat, healthy kind.

5. Real silicon results (1,439-die dataset)

Metric	CDYN V1 (0.7 V)	CDYN V2 (0.95 V)
Median (typical die)	31.29 nF	32.03 nF
Minimum	30.30 nF	28.04 nF (low flyer)
Maximum	31.90 nF	32.56 nF

V1→V2 median rises only **+2.4%** — the healthy, stable signature. CDYN is essentially flat vs. UPM (speed), meaning dynamic capacitance is not what sorts die speed — expected, and a good sign the extraction is clean. One low die at 28.04 nF (V2) is the only follow-up item.

6. Verification — are the numbers correct?

I recomputed CDYN from the raw DICC for every die and compared to what the tester logged:

Check	Result
Dies checked	1,439 (all valid)
V1 matches tester (<0.001 nF)	1,439 / 1,439
V2 matches tester (<0.001 nF)	1,439 / 1,439
Largest disagreement	0.0008 nF (< 1 pF)
Dies that disagreed	0

Bottom line: the CDYN arithmetic is proven correct on real silicon (1439/1439). The remaining layers to watch are the raw DICC measurement quality (fit linearity, monotonicity) and the absolute level vs. spec — noted as next-step checks.

Appendix A. Device Ce_{eff} from e-test (RA4 / RJ4 / RK4)

E-test (PCM/WAT) does not measure Ce_{eff} directly; it is derived from ring-oscillator (RO) structures using the gate-delay relation. Each RO family (RA4, RJ4, RK4 — different device constructs) has its own matched drive current and delay:

$$C_{\text{eff}} = I_{\text{eff}} \times T_d \div V_{\text{dd}}, \quad I_{\text{eff}} = (|I_{\text{sat,N}}| + |I_{\text{sat,P}}|) \div 2$$

Per-family medians ('u' flavor, V_{dd} = 0.75 V):

Family	I _{eff} (mA/μm)	T _d (ps)	C _{eff} (fF/μm)	Reticle spread
RA4 (main)	5.579	6.564	48.76	11.5%
RJ4	4.759	5.878	36.90	7.7%
RK4	4.539	5.536	33.72	7.0%

Units — the key subtlety

- **Spec unit is μA/μm** (e.g., Isat_RNJ4S target ~2947 μA/μm). The stored data is in **mA/μm** (spec ÷ 1000): data 3.278 × 1000 = 3278 μA/μm ≈ spec.
- With Isat in mA/μm and T_d in ps, **C_{eff} comes out in fF/μm** — a per-micron density.
- **T_d → frequency**: with T_d in ps and divide-by-2, RO freq = 1/(2·T_d). UPM = 500/T_d is that frequency; RA4u ≈ **7,600 MHz (~7.6 GHz)**. Higher UPM = faster = lower Ce_{eff}.

Appendix B. CDYN vs device Ceff — correlation study

Goal: test whether device RA4u Ceff (e-test) predicts GT sort CDYN. Challenge: only 9 e-test sites vs. 1,439 sort dies, so we spatially interpolate the 9 sites onto every die using Inverse-Distance Weighting (IDW), then correlate.

IDW method

For each die, estimate Ceff as a distance-weighted average of the 9 sites: closer sites get more weight (weight = $1/\text{distance}^p$, $p=2$). The 9 sites are placed at their die-coordinate centroids from the reticle map (4 dies per reticle shot).

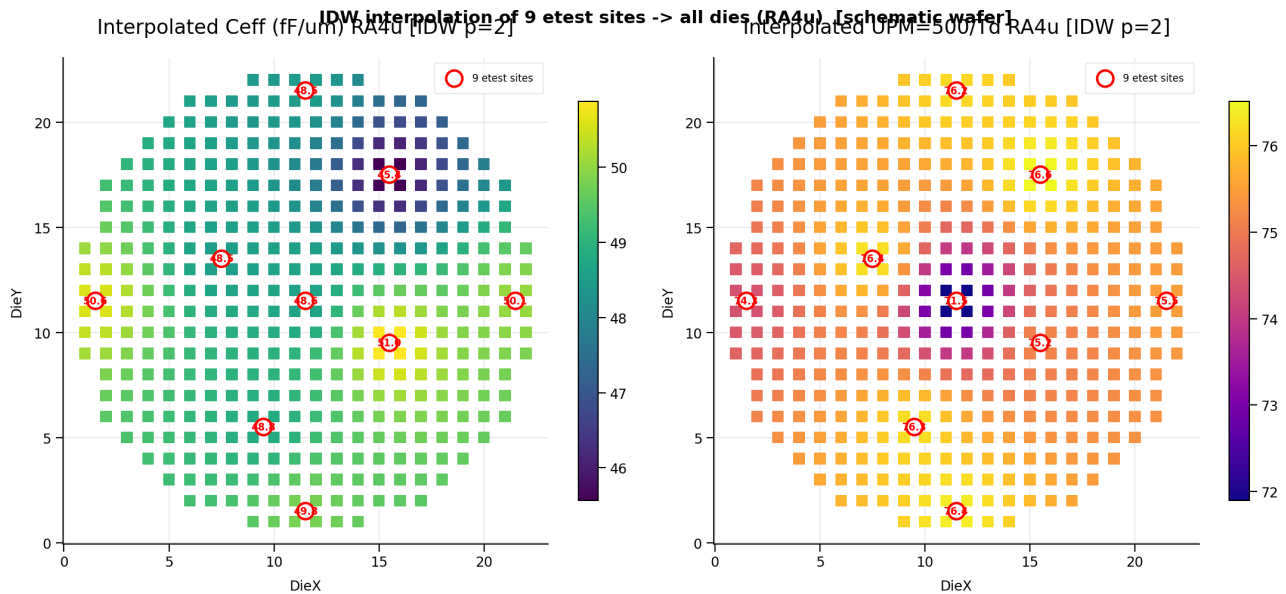


Figure B1. IDW interpolation of the 9 e-test sites onto all dies (schematic wafer). Red circles = the 9 measured sites; low-Ceff / high-UPM pocket near X2Y3.

Correlation result (1,439 dies, 4 wafers)

Correlation	CDYN_V1 (0.7 V)	CDYN_V2 (0.95 V)
Pearson r	+0.049	+0.069
Spearman ρ	+0.134	+0.168
Slope (nF per fF/ μm)	0.012	0.019
Variance explained (r^2)	~0.2%	~0.5%

Conclusion: GT CDYN is BEOL/wire-limited, not device-limited

Pearson $r \approx 0.05$ – 0.07 is effectively zero — RA4u device Ceff explains $<0.5\%$ of CDYN variance. This is **why device Ceff (~49 fF/ μm) never matched sort CDYN (~31 nF)**: they are different capacitances. RA4u is a FEOL device speedometer; GT CDYN is a full-stack, interconnect-dominated product quantity (device + BEOL wire + clock tree + arrays).

Practical implication: no single e-test parameter — FEOL Isat or even BEOL Con_M — will reproduce absolute CDYN; at best it gives a **trend** for process control / excursion screening. The slightly higher Spearman ρ hints at a faint monotonic (likely radial) commonality. Next step: correlate CDYN against BEOL

PCM (Con_M*, SPA_M*, via Rc_V*) at the site level (no IDW smearing) to find which layer carries the strongest trend.

Key relationships (quick reference)

Quantity	Formula	Typical value
CDYN	$1000 \times \text{slope}(\text{DICC vs } f) \div V_{cc}$	31 nF (V1), 32 nF (V2)
Device Ceff (RA4u)	$I_{eff} \times T_d \div V_{dd}$	48.8 fF/ μm
RO frequency	$1 \div (2 \times T_d)$	~7.6 GHz
UPM	$500 \div T_d[\text{ps}]$	~76
CDYN vs Ceff	Pearson r	~-0.05 (uncorrelated)